

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels were produced by MTurk workers explicitly restricted to 'countries where native English speakers are most likely to be found' **[Q61]**, with majority vote across three workers per timeline **[Q70]** and reference summaries selected from the two highest-agreeing workers **[Q88]**. Summary quality ratings used five MTurk workers under the same location restriction **[Q105, Q106]**. The deployment requires labels produced by an inter-operational team of Ecuadorian field coordinators, social analysts, and domain experts (elicitation A3) — a fundamentally different and unavailable annotator population. There is no Spanish-speaking, Latin American, indigenous-language-fluent, or humanitarian-practitioner annotator in the entire benchmark pipeline. The high inter-annotator agreement (90.06% **[Q77]**) demonstrates only internal consistency among US-context workers, not validity for the Ecuadorian humanitarian construct.

ID	Evidence
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q70	"To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers." (p.6)
Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q88	"To adapt the timeline summarization task to this setting, we pick the two summaries written by the two workers who agree the most based on the timeline extraction labels they assign to the tweets in each timeline." (p.7)
Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)

Strengths

- The benchmark documents annotation procedures transparently — multiple QC gates **[Q60–Q66]**, qualification testing **[Q62]**, expert review **[Q64]**, and reported agreement statistics **[Q77, Q79]** — providing a methodological template that could be adapted (with different annotators) for Ecuadorian re-annotation.

ID	Evidence
Q60	"Following prior quality control practices (Briakou et al., 2021), we use multiple quality control (QC) steps to ensure the recruitment of high-quality annotators." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q62	"Next, we recruit annotators who pass our qualification test (QC2), where we ask them to annotate 3 timelines." (p.5)
Q63	"We run several small pilot tasks, each with a replication factor of nine." (p.5)
Q64	"We check annotators' performance on timeline extraction task against experts' labels and have experts manually review (QC3) annotators' summary quality." (p.5)
Q65	"Only workers passing all the quality control steps contribute to the final task." (p.5)
Q66	"During the final task, we perform regular quality checks (QC4), and only use workers who consistently perform well." (p.5)

Q77	"The average timeline-level agreement between those two workers is 90.06%." (p.6)
Q79	"The average timeline-level agreement rate of this analysis is 91.77%." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — labels reflect English-native MTurk worker judgments [Q61, Q70], not Ecuadorian humanitarian coordinator perspectives.

OC-2: Disagreement is highly likely on the 'informative' criterion [Q58] (excluding personal views vs. needs-signal interpretation), on what constitutes 'important information' for Coverage [Q157], and on event boundaries (US local-fire model vs. multi-phase volcanic crisis).

OC-3: Demographic documentation is partial: only the location restriction [Q61], MTurk recruitment, qualification process [Q62–Q66], and pay rate [Q67–Q68] are documented. No information on language background beyond English-native, professional background, or humanitarian experience.

OC-4: Re-annotation by an Ecuadorian inter-operational team (per A3) is essential — but the existing labels themselves are not reusable as a starting point for humanitarian relevance.

OC-5: Aggregation is majority vote of three workers [Q70], which can erase a single dissenting humanitarian-aware perspective — particularly risky for low-frequency but high-importance signals (community needs, indigenous voices).

OC-6: Convergent validity and external validity are both violated: there is no evidence the labels correlate with Ecuadorian humanitarian judgments, and the annotator population is operationally unrepresentative.

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q61	"First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)
Q62	"Next, we recruit annotators who pass our qualification test (QC2), where we ask them to annotate 3 timelines." (p.5)
Q63	"We run several small pilot tasks, each with a replication factor of nine." (p.5)
Q64	"We check annotators' performance on timeline extraction task against experts' labels and have experts manually review (QC3) annotators' summary quality." (p.5)
Q65	"Only workers passing all the quality control steps contribute to the final task." (p.5)
Q66	"During the final task, we perform regular quality checks (QC4), and only use workers who consistently perform well." (p.5)
Q67	"We compensate the workers at a rate of \$3 per HIT for the task." (p.5)
Q68	"Each batch of tasks is followed by a one-time bonus that makes the final rate over \$10 per hour." (p.5)
Q70	"To make the ground-truth on whether each tweet is part of the timeline or not, we take the majority label among the three workers." (p.6)

Q105	"Following a similar process in 3.2, we recruit a group of workers from MTurk with the same Location restriction (QC1), who can pass our pilot study as a qualification test." (p.8)
Q157	"For each summary, answer the question "How well does the summary cover the important information in the timeline?" on a scale of 1 to 5. A good summary has good coverage if it mentions the main information from the timeline that's important to understand the event. A summary has poor coverage if someone reading only the summary would be missing several important pieces of information about the event in the timeline." (p.21)
WEB-22	HumAID 2016 Ecuador Earthquake subset shows severe class imbalance (e.g., 3 tweets in 'displaced people and evacuations'), illustrating that even Ecuador-event data annotated under generic schemas underweights humanitarian categories. (https://www.researchgate.net/publication/350720422_HumAID_Human-Annotated_Disaster_Incidents_Data_from_Twitter)

Information Gaps

- Specific country-level distribution of the MTurk worker pool (US-only? US/UK/AU?).

Requires Expert Verification

- Whether any Cruz Roja Ecuatoriana or UN OCHA Ecuador annotation guidelines could anchor a re-annotation effort.

Remediation

Gap 1: Labels and reference summaries produced by English-native MTurk workers, not by the Ecuadorian inter-operational team that will use the system.

Recommendation: Conduct re-annotation with field coordinators, social analysts, and domain experts from Cruz Roja Ecuatoriana and UN OCHA Ecuador; use deliberative consensus rather than majority-vote aggregation to preserve minority humanitarian signals; document annotator demographics in a Datasheet.